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## Mesenchymal stem cell-derived secretome and its applications

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### Abstract

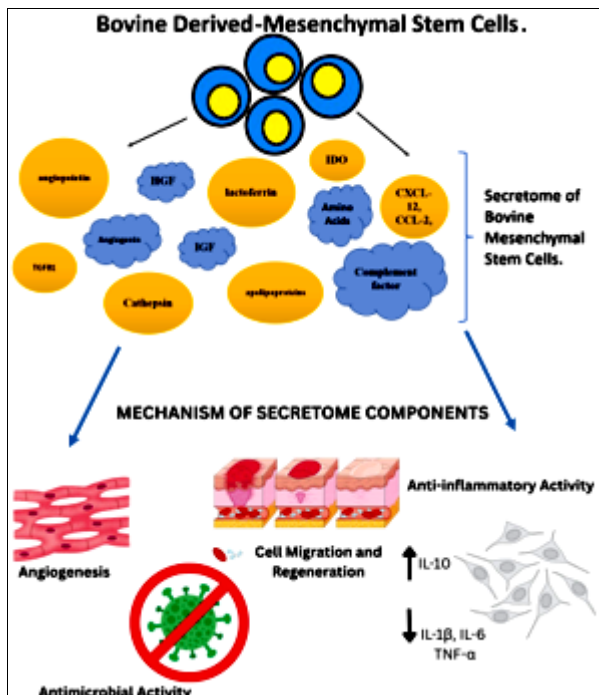
Secretome derived from mesenchymal stem cells is an alternative approach for regenerative medicine without involving cells directly in the applications. The bovine mesenchymal stem cells secretome (bMSC) consists of bioactive molecules such as growth factors, antimicrobial peptides, chemokines, extracellular vesicles, metabolites and amino acids. The MSC secretome content varies depending on the bovine tissue it is sourced from. The potential of the secretome in various applications was studied and proved effective for reducing relapses for mastitis, reducing inflammation in the endometritis model, re-epithelialisation in wound healing and enhancing lipogenesis in mammary epithelial cells for milk production. The secretome also showcases anti-inflammatory and anti-apoptotic properties. The secretome application has more advantages over direct stem cell therapy as it exhibits low immunogenicity easy to store and has no risk of teratoma formation. This review explores the therapeutic potential of the bMSC secretome and its significance in treating cattle diseases.

**Keywords:** Bovine mesenchymal stem cells (bMSC), secretome, growth factors, antimicrobial, anti-inflammatory, anti-apoptotic

### 1. Introduction

The secretome derived from bovine mesenchymal stem cells (bMSCs) offers a novel cell-free therapeutic approach with great potential and as a solution for economically important diseases in the cattle industry. The secretome derived from bovine MSCs comprises important bioactive molecules, including growth factors, cytokines, chemokines, secondary metabolites, amino acids and extracellular vesicles, which can enhance cellular migration and immune activity in conditions such as inflammation and wound healing. The secretome therapy approach has many added advantages over direct MSCs transplantation in regenerative medicine, including reduced risk of tumour formation (teratoma), minimal immunogenicity, simplified clinical administration and ease of preservation and storage under frozen conditions. For acute conditions such as fulminant mastitis, acute reproductive disorders and traumatic injuries, the secretome derived from bovine MSCs is readily available, making it especially valuable for time-sensitive interventions<sup>[1-3]</sup>.

The production aspect of the secretome involves cell isolation, expansion and the collection of conditioned media from the cells, which contains the secretome derived from them. This straightforward method enables the production and collection of the growth factors directly from MSCs without the need for costly expression systems. Overall, this approach to harvesting bioactive molecules is both efficient and cost-effective.



**Fig 1:** Graphical illustration of the Bovine-Derived Mesenchymal Stem Cells Secretome components and mechanism

## 2. Secretome composition and characteristics of the bioactive compounds

### 2.1 Growth factors

#### Cytokines and Angiogenic Factors

The secretome derived from bovine mesenchymal stem cells (MSCs) exhibits a diverse protein composition that varies depending on the tissue source and given stimulation conditions. A proteomic study of the secretome derived from bovine endometrial mesenchymal stem cells (eMSCs) revealed 302 unique proteins, which were expressed under control conditions and 397 proteins following stimulation with lipopolysaccharide (LPS). Among the proteins found in the LPS-stimulated group, transforming growth factor beta (TGFB1) was identified, which is known to be a potent growth factor. Notably, 242 protein groups were found to be common to both the control-derived secretome and the LPS-stimulated-derived secretome. These bioactive compounds span multiple functional categories, including cell migration, angiogenesis, immune stimulation and tissue remodelling. Type IV collagenase, leucine-rich alpha-2-glycoprotein 1 and angiogenin-2 are key factors identified in the secretome that facilitate angiogenesis. The bioactive molecules expressed by eMSCs are essential to the intricate process of angiogenesis, highlighting their importance in physiological contexts of blood vessel formation [4].

Bovine mammosphere-derived cells (MDCs) and adherent fraction-derived cells (AFDCs) are two cell populations that were derived from bovine mammary gland tissue. The non-adherent MDCs were derived from the adherent AFDCs cell population. Despite the common tissue origin, these two cell populations exhibit variations in the secretome, yet also show some overlap of the secreted proteins. The liquid chromatography-Mass spectrometry analysis has identified a total of 347 proteins in the MDC-derived secretome and 537 proteins in the AFDC-derived secretome. Fibroblast growth factor (FGF), hepatocyte growth factor (HGF), vascular endothelial growth factor A (VEGFa), angiopoietin-1 (Ang-1) and transforming growth factor beta (TGFβ) are important

factors identified in the secretome that enhance cellular migration and angiogenesis [5].

The secretome derived from both MDCs and AFDCs contains a diverse array of angiogenic factors that are critically important for tissue repair. VEGF is recognised as a primary angiogenic molecule, with the AFDCs secretome exhibiting significantly higher levels of the VEGFa compared to that of the MDCs secretome. Angiopoietin-1 is a crucial protein in the maintenance and development of blood vessels. This particular factor is found to be abundant in the MDCs-derived secretome, when quantified using enzyme-linked immunosorbent assay (ELISA) [6]. These factors interact synergistically to enhance the formation and stabilisation of the new blood vessels. This shows both the secretome's unique and connected roles in supporting angiogenesis and enhancing tissue repair.

HGF and IGF-1 are recognised as important growth factors and have significant roles in cell migration and proliferation. Notably, these factors are expressed in the secretome of both MDCs and AFDCs. An investigation revealed that Madin-Darby Bovine Kidney cells (MDBK) cultured in the presence of secretomes derived from MDCs and AFDCs exhibit the expression of receptors relevant to the growth factors present in the secretomes. The receptors IGF receptor, MET proto-oncogene receptor kinase and TGF beta receptor for IGF, HGF and TGF-β growth factors were expressed. The upregulation of the relevant receptor expression after exposure to the growth factors containing the secretome underscores the functional importance of the secretome in cell signalling, regeneration and tissue growth [7].

Analysis of the secretome derived from bovine umbilical cord mesenchymal stem cells (BUMSCs) demonstrates a diverse composition of bioactive components that contribute to as a therapeutic approach. The key growth factors, which include platelet-derived growth factor (PDGF), transforming growth factor (TGF), epidermal growth factor (EGF) and hepatocyte growth factor (HGF) are crucial growth factors in promoting cell proliferation and tissue repair and VEGF for angiogenesis. Stromal-derived factor 1 alpha (SDF-1a) supports cell migration to the site of injury and enhances stem cell proliferation and differentiation [8]. The secretome also contains chemokine receptors such as CXCR4 and CCR4, as well as many adhesion molecules, that includes VCAM, ICAM and PECAM. Additionally, chemokines such as CCL-3, CCL-2 and CXCL-12 are present and essential in cellular signalling and communication. Metalloproteinases like ADAM-12 further contribute in cell migration. These components collectively underscore the potential of the BUMSC secretome in regenerative therapy applications [9].

### 2.2 Amino Acid Profile

The metabolomic analysis of bovine umbilical cord mesenchymal stem cells (BUMSCs) secretome identified numerous amino acids Alanine (Ala), Arginine (Arg), Glutamate (Glu), Glutamine (Gln), Histidine (His), Isoleucine (Ile), Leucine (Leu), Methionine (Meth), Phenylalanine (PheAla), Threonine (Thr), Tryptophan (Try), and Tyrosine (Tyr). Other than amino acids, metabolites, and vitamins, including acetate, choline, lactate, α-glucose, β-glucose, pyruvate, and nicotinamide (Kusindarta and Wihadmadyatami, 2021). The metabolomic profile of bovine umbilical vein endothelial cell-derived (BUVECs) secretome reveals a comparable amino acid and secondary metabolite composition to that of BUMSCs secretome. Additionally, the

secretome contains vitamins such as cholecalciferol and nicotinamide [10].

### 2.3 Antimicrobial Peptides and Defense Proteins

The bovine milk-derived cell secretome demonstrates substantial antimicrobial properties. Proteomic analysis has revealed the presence of several antimicrobial peptides, including lactoferrin, pantetheinase, cathepsin, mannose-binding protein C, beth-2-microglobulin and protein S100-A11, found in both MDCs and AFDCs secretome. These proteins act as inhibitors of bacterial growth and as innate immune factors [7]. This evidence proves the secretome's potential for combating infections and improving bovine health.

The proteomic profile of eMSCs-derived secretome revealed an enrichment of antibacterial response proteins and macrophage activation proteins, particularly when the eMSCs

were stimulated with LPS. The increased protein expression and diversity following LPS stimulation enhance the antimicrobial and immunomodulatory components of the secretome, which contribute to tissue remodelling. Interestingly, even under unstimulated control conditions, the eMSCs secretome contained proteins with antimicrobial properties. Some of the identified antibacterial and antifungal proteins are serpin, cystatin, neutrophil defensin-1 and apolipoprotein A-II. LPS stimulation has also been observed to facilitate anti-inflammatory mediator expression, such as granulins in eMSCs. The presence of anti-inflammatory proteins in the secretome indicates that the environmental condition will influence the bioactive components that are secreted by eMSCs [4]. Antimicrobial peptides, which include complement factor B, complement component C9, lactotransferrin and apolipoproteins, were found to be present in the secretome derived from amniotic progenitor cells [11].

**Table 1:** Secretome from various bovine tissue sources

| Si. No. | Tissue source of the bovine mesenchymal stem cells.            | Contents of the secretome                                                                                                                                                                        | Biological functions                                                                                                                                        | Proteomic analysis for identification of bioactive factors in secretome | References |
|---------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------|
| 1.      | Milk-derived progenitor cells                                  |                                                                                                                                                                                                  | Antimicrobial responses and cell migration properties.                                                                                                      |                                                                         | [5]        |
| 2.      | Mammary epithelial cells                                       |                                                                                                                                                                                                  | Increased phagocytosis in macrophages                                                                                                                       |                                                                         | [14]       |
| 3.      | Mammosphere-derived cells from mammary epithelial cells        | Alpha-2-macroglobulin, Cathepsin, angiopoietin-related protein 3 precursor, vascular endothelial growth factor A isoform precursor, hepatocyte growth factor and insulin-like growth factor etc. | Role in immune defense, angiogenesis and cellular migration.                                                                                                | Mass spectrometry and ELISA                                             | [7]        |
| 4.      | Adherent fraction-derived cells from mammary epithelial cells. | Alpha-2-macroglobulin, Cathepsin, phospholipid transfer protein, angiopoietin-related protein 3 precursor, hepatocyte growth factor and insulin like growth factor etc.                          | Role in immune defense, angiogenesis and cellular migration.                                                                                                | Mass spectrometry and ELISA                                             | [7]        |
| 5.      | Umbilical mesenchymal stem cells                               | Growth factors, amino acids and vitamins.                                                                                                                                                        | Neuroprotective property                                                                                                                                    | LC-MS/MS                                                                | [9]        |
| 6.      | Umbilical vein endothelial cells (BUEVCs)                      | a) Vascular endothelial growth factor and basic fibroblast growth factor.                                                                                                                        | Angiogenesis and cell migration properties.                                                                                                                 | ELISA                                                                   | [19]       |
|         |                                                                | b) Amino acids, secondary metabolites and vitamins.                                                                                                                                              | Neuroprotective, anti-apoptotic and anti-inflammatory properties.                                                                                           | LC-MS                                                                   | [10]       |
| 7.      | Endometrial mesenchymal stem cells                             | Annexin A1, Neutrophil defensin 1, Caspase 14, Granulins, Plasminogen, Angiogenin-2, Heat shock 27 kDa protein, peroxiredoxin-6, fibrinogen alpha chain, collagen alpha chain.                   | Immune response, antimicrobial activity, Anti-inflammatory activity, angiogenic activity, tissue remodelling, protection from stress and cellular adhesion. | LC-MS/MS                                                                | [4]        |
| 8.      | Wharton's jelly layer-derived mesenchymal stem cells           | -                                                                                                                                                                                                | Anti-inflammatory activity and enhanced lipogenesis in Mammary epithelial cells.                                                                            | -                                                                       | [18]       |
| 9.      | Amniotic cells                                                 | Complement component C9, Complement factor B, Insulin-like growth factor-binding protein 5, Lactotransferrin.                                                                                    | Growth factors for regenerative activity, antimicrobial activity and immune defence.                                                                        | LC-MS                                                                   | [11]       |

### 3. Mechanism of Action

#### 3.1 Immunomodulatory Properties

The secreted proteins from bovine adipose-derived mesenchymal stem cells (bAD-MSCs) include significant anti-inflammatory proteins. In models of LPS-induced inflammation, the co-culture of bAD-MSCs with bovine endometrial epithelial cells (bEECs) leads to a marked reduction in pro-inflammatory cytokines. This interaction between the secretions of the bAD-MSCs and the bEECs highlights the potential of the bAD-MSCs secretome for managing inflammation-related conditions in cattle and as a therapy option based on the secretome's role in modulating immune response and promoting a healthier state in affected

tissues. Specifically, treatment leads to significant decreases in the expression levels of interleukin-1 $\beta$  (IL-1 $\beta$ ), interleukin-6 (IL-6), and tumour necrosis factor-alpha (TNF- $\alpha$ ) in inflamed endometrial epithelial cells. At the molecular level, the secretome of bAD-MSCs downregulates p38 phosphorylation, inhibitor of kappa B alpha (I $\kappa$ B- $\alpha$ ) expression, and Janus kinase 1 (JAK1) phosphorylation induced by LPS—key signalling molecules involved in inflammatory responses. The bAD-MSCs-derived secretome has been observed to downregulate the expression and phosphorylation of apoptotic proteins such as Bax and p38 mitogen-activated protein kinase [12]. Such mechanisms enhance the cell survival capacity under stress conditions.

Moreover, bAD-MSCs secrete substantial amounts of indoleamine 2,3-dioxygenase (IDO) under both unstimulated and interferon-gamma (INF- $\gamma$ ) stimulated conditions, demonstrating the immunomodulatory capacity in both naïve and activated conditions. IDO prevents autoimmunity and exhibits immunosuppression by the regulation of T effector and T regulatory cells<sup>[13]</sup>.

Similar to umbilical blood cord stem cells, bovine umbilical vein endothelial cells can also be applied to secretome therapy. Larasati *et al.* (2022) studied the secretome derived from bovine umbilical vein endothelial cells (BUVECs) for its anti-inflammatory and anti-apoptotic effects. As the secretome contained many metabolites, which contained many amino acids and vitamins. The TMT-induced apoptotic and inflammatory effects were downregulated in the presence of the secretome in PC12 and SH-SY5Y cell lines. This showcases the neuroprotective effects of the secretome.

The immunomodulatory effects of bovine mammosphere-derived epithelial cells (MDEC) on macrophages were studied by Danev *et al.*<sup>[14]</sup>. MDEC is a stem cell-enriched culture that secretes many immunomodulatory factors that enhance phagocytosis. This was confirmed when the M1 and M2 macrophage phenotypes were co-cultured with pH-sensitive bioparticles and secretome derived from MDEC, increased phagocytosis of the particles was observed by the mean fluorescent intensity of the macrophages. A notable increase in the ROS accumulation in both the macrophage phenotypes was observed when cultured in the presence of MDEC conditioned media (secretome). This study suggests the MDEC-derived secretome enhances phagocytosis in macrophages, which is an essential first line of defence against pathogen entry.

Extracellular vesicles are a major part of the secretome, which also contains bioactive compounds with a significant role. A comparative study was conducted to evaluate the effectiveness of antibiotic treatment versus extracellular vesicles derived from bovine umbilical cord-derived mesenchymal stem cells (UCB-MSC-EVs) in treating subclinical mastitis in dairy cows. The results indicated that the groups treated with UCB-MSC-EVs showed a significantly greater effect compared to those receiving antibiotics. Upregulation of anti-inflammatory cytokine (*IL-10*) expression in the EV-treated group was upto 25 fold increase, and a reduced somatic cell count in milk was observed in the UCB-MSC-EV-treated groups rather than the antibiotic-treated groups<sup>[15]</sup>.

### 3.2 Angiogenic Activity

Secretome derived from bovine mammosphere-derived cells promotes angiogenesis by activating endothelial cells. In angiogenesis studies conducted by Ledet *et al.*<sup>[7]</sup>, the MDC and AFDC-derived secretome stimulate the bovine microvessel endothelial cells (BLMVEC) tube formation, which demonstrates the pro-angiogenic potential of the secretome. This activity correlates with VEGFa and Ang-1 content, and the mechanism is independent of cell proliferation, suggesting direct activation of angiogenic signalling pathways in endothelial cells. The secretome derived from the MDC was observed to increase the mean tube length of the endothelial cells compared to the control maintained in DMEM. This can be possible due to the increased presence of angiopoietin 1. Increased numbers of endothelial cells were observed when cultured in AFDC secretome, which has higher levels of VEGFa.

### 3.3 Cell Migration and Epithelial Regeneration

Bovine MSCs-derived secretome promotes epithelial cell migration through multiple components present in the secretome. In *in vitro* scratch assays, the MDC secretome enhanced MDBK cells migration compared to the control medium. The observed effect is due to the presence of the crucial growth factors such as IGF-1, HGF and TGF $\beta$ . The antibody inhibition studies reveal that inhibiting the action of HGF with anti-HGF antibody alone significantly reduces the cell migration<sup>[7]</sup>. Secretome derived from bovine umbilical vein endothelial cells exhibits re-epithelialization and fibroblast regeneration when applied in burn wounds and has also aided in the closure of the burn wounds.

### 3.4 Antimicrobial Activity

Amniotic progenitor cell-derived conditioned medium (CM) has exhibited a notable capacity to attenuate bacterial growth *in vitro*. When the bovine mammary epithelial cells (BMEs) are co-cultured with *Staphylococcus aureus* in the presence of 10% secretome derived from bovine amniotic progenitor cells, bacteriostatic effect and anti-apoptotic effect were observed with a viability rate of 89% after 24 hours of incubation with *Staphylococcus aureus*. Further *in vivo* studies on secretome treatment on dairy cows diagnosed with acute and chronic mastitis were carried out. This resulted in a reduction of relapses associated with chronic mastitis when compared to the groups treated with antibiotics. This suggests a particular efficacy of CM in addressing recurrent infections, which may be attributed to its constituent factors, including lactoferrin, albumin, and prothrombin proteins<sup>[11]</sup>.

Additionally, the antimicrobial properties of the secretome derived from milk-derived cells (MiDC) have been documented to inhibit the growth of mastitis-associated pathogens such as *Klebsiella pneumoniae*, *Escherichia coli*, *Streptococcus uberis*, as well as *Methicillin-resistant Staphylococcus aureus* (MRSA), a strain known for its multidrug resistance. The observed antibacterial effects against these pathogens present significant implications for the treatment of mastitis<sup>[5]</sup>.

The expression of antimicrobial peptides (*CATHL4*, *LCN2* and *CST3*), which are a part of the innate immune system, was observed to be elevated after the treatment with UCB-MSC-EV in subclinical mastitis conditions<sup>[15]</sup>.

## 4. Collection and Preparation Methods

### 4.1 Culture Conditions

Standard protocols involve culturing MSCs in serum-free medium for 24-96 hours to collect conditioned medium (CM). Optimisation of culture conditions is critical for ensuring the biological activity of the secretome. The addition of Fetal bovine serum (FBS) to the MSCs culture for secretome induction is avoided. This is to prevent the interference of the serum proteins with the secretome proteins and the identification of secretome proteins alone. The cell density in the culture dish should be 70%-80% confluent. This is to enable the cells to proliferate in the absence of FBS and to facilitate the secretion of growth factors in the absence of FBS.

### 5. Yield and Scalability

Bovine tissue sources provide substantial quantities of secretome suitable for clinical application. Approximately 120 liters of conditioned medium can be obtained from a single umbilical cord-derived Wharton's jelly mesenchymal

stem cells when standardised collection protocols upto fourth passage at a cell seeding concentration of 4000 cells per cm<sup>2</sup> are employed, facilitating commercial production. This scalability distinguishes bovine sources from smaller animal models and enables industrial-scale manufacturing [2].

For better yield of secretome and expansion of MSCs, microcarrier culture with stirred tank bioreactor systems and multi-layered cell culture flasks can be employed. Comparing the expansion of cells in a fixed area, such as culture flasks, suspension cultures will allow expansion of a higher density of cells in less space. A maximum of 1000 liters of conditioned medium can be harvested from a single batch of suspension microcarrier culture [16,17].

## 6. Therapeutic Application of Bovine Mesenchymal Stem Cell-Derived Secretome

### 6.1 Bovine Mastitis, Endometritis and the Clinical Efficacy of Secretome

Mastitis in dairy cattle is the most economically damaging disease globally, as it can reduce milk production efficiency. This inflammatory condition extensively damages udder tissues and the full restoration of milk production capacity cannot be achieved with current therapeutic strategies. The composition of the bovine secretome derived from mesenchymal stem cells (MSCs) exhibits a diverse array of functions that play a crucial role in mitigating inflammation within the udder region. It demonstrates remarkable bacteriostatic properties, effectively inhibiting bacterial growth, while also possessing potent regenerative capabilities that promote healing, angiogenesis and recovery.

Amniotic-derived conditioned medium (secretome) demonstrates therapeutic potential in both acute and chronic mastitis. A clinical study was conducted involving 48 dairy cows with acute and chronic mastitis. The secretome treatment reduced the bacterial growth *in vitro* and significantly reduced relapses in chronic mastitis compared to the antibiotic treatment group. This significant outcome in chronic case treatment suggests that the secretome holds regenerative and immune-modulating properties that address the underlying tissue damage in the udder rather than suppressing acute inflammation. The extracellular vesicles secreted by mesenchymal stem cells derived from bovine umbilical cord blood have demonstrated their efficacy in mastitis treatment. These vesicles are more effective than antibiotic treatments, as they influence gene expression related to anti-inflammatory processes and tissue regeneration. The inflammation in the mastitis condition is well downregulated when treated with the secretome [11, 5, 15]. The inflammation in the mastitis condition can also be downregulated when treated with the secretome. Bovine adipose-derived MSC secretome effectively suppresses LPS-induced inflammation in bovine endometrial epithelial cells *in vitro*. The bAD-MSCs secretome treatment produces a dose-dependent reduction in pro-inflammatory cytokines in the LPS-stimulated bEECs. The expression of cytokines IL-1 $\beta$ , TNF- $\alpha$  and IL-6 in bEECs was reduced to baseline or near-baseline levels. The secretome also suppresses the JAK-STAT signalling and the I $\kappa$ B- $\alpha$  protein expression, which are important in inflammatory gene expression [12].

Co-culturing of mammary epithelial cells with Wharton's jelly-derived mesenchymal stem cells has also significantly downregulated interleukin-6 and tumour necrosis factor  $\alpha$  [18].

### 6.2 Mechanisms of Protection against Bacterial Infection

The secretome treatment protects the bovine mammary

epithelial cells from the LPS-induced inflammation and injury through multiple complementary mechanisms. In co-culture models, bAD-MSC secretome was observed to significantly reduce the LPS-induced apoptosis in bEECs, suggesting similar protective effects in mammary tissue. The protective nature of the MDC and AFDC secretome towards the LPS-treated bovine epithelial cells was also documented. The bovine eMSCs were observed to secrete antimicrobial peptides when stimulated with LPS. This proves that bovine MSCs of any tissue source secrete antimicrobial factors also in the presence of pathogens as a protective mechanism. The secretome's antimicrobial component, including lactoferrin, apolipoprotein A-II, neutrophil defensin-1 and cathepsin, provides direct antibacterial activity [12, 4, 7].

### 6.3 Milk Production Enhancement and Stress Response

Beyond treating pathological mastitis, the bovine MSC secretome may enhance milk production in healthy mammary glands. Co-culture studies demonstrate that MSC secretome significantly increases expression of lipogenic genes in mammary epithelial cells, including fatty acid synthase (FASN), acetyl-CoA carboxylase (ACC) and increased TAG, suggesting potential to enhance milk fat synthesis. The secretome has also demonstrated modulation in stress response and elevated expression of lipogenesis in bovine mammary epithelial cells (MEC) after exposure to LPS as a stress factor. The expression of FASN and ACC has significantly increased in the LPS-treated MEC in the presence of the secretome. This study suggests that secretome application can improve the mammary gland's resilience to metabolic and inflammatory stressors during peak lactation periods [18].

### 6.4 Wound Healing and Tissue Regeneration

Bovine MSCs secretome demonstrates substantial therapeutic potential for accelerating wound closure across multiple wound types.

Bovine umbilical vein endothelial cell (BUVEC) secretome promotes burn wound healing through multiple coordinated mechanisms. The topical application of BUVEC-CM to the burn wounds promotes fibroblast regeneration, collagen formation, re-epithelialization and hair follicle regeneration. On the 5<sup>th</sup> day of post-treatment, the wound closure was observed with significantly enhanced epithelial repair by day 13 of post-treatment. Collagen and the formation of new blood vessels were substantially increased by day 21 and 29 of post-treatment [19].

The pro-angiogenic nature of secretome is due to the presence of components (VEGF, Ang-1), which promote new blood vessel formation essential for wound healing. Simultaneously, migration-promoting factors (HGF, TGF $\beta$ , IGF-1) stimulate fibroblast migration and proliferation, enhancing extracellular matrix deposition [7].

## 7. Advantages of Cell-Free Secretome Therapy

### 7.1 Safety Profile

The cell-free nature of secretome eliminates risks associated with direct MSC transplantation, including tumour formation (teratomas), cellular rejection, and pulmonary embolism from intravenous cell administration. The benefits of the MSC therapy can be attained even from exosomes derived from MSCs, which can also be used to complement other treatments. The primary concern of FBS usage is the risk of zoonotic disease transmission, which can be prevented in the secretome application, as FBS is not required for the

secretome harvest from the respective cells <sup>[1, 20]</sup>.

## 7.2 Manufacturing and Scalability

Unlike living cell products requiring sophisticated cryopreservation and thaw protocols, secretome can be concentrated, lyophilised, packaged and transported with relative ease. Approximate yields of 120 liters per umbilical cord provide suitable quantities for industrial-scale production. The storage of secretome does not require any cryoprotectant and can be easily stored at -80 °C or lyophilised, stored and easily transported <sup>[1, 2]</sup>.

## 7.3 Immediate Availability

For acute conditions, including mastitis, acute reproductive crises, and traumatic injuries, the pre-manufactured secretome, which was stored previously, is immediately available without requiring time-consuming cell isolation and expansion.

## 7.4 Standardisation Potential

Secretome composition can be characterised similarly to conventional pharmaceutical agents, with proteomic profiling enabling batch-to-batch comparison and quality control. Large-scale production of secretome using suspension microcarrier bioreactors can be standardised, similar to other pharmaceutical bioprocesses <sup>[17]</sup>.

## 8. Current Limitations and Regulatory Challenges

### 8.1 Incomplete Mechanistic Understanding

The secretome studies provide evidence of efficacy across various applications. The relative contribution of EV-mediated versus soluble factors-mediated effects remains incompletely characterised and documented. Some studies suggest that application of the whole secretome containing both fractions provides more effective results than EV treatment alone. The specific mechanisms underlying this synergistic action of bioactive compounds require further elucidation <sup>[21, 22]</sup>.

### 8.2 Limited Clinical Data

Despite promising *in vitro* and limited *in vivo* results, systematic clinical trials comparing bovine secretome efficacy to current standard veterinary treatments remain sparse. Most published clinical applications involve mastitis conditions and expansion of clinical trials on other conditions, such as dermal wounds and other inflammatory conditions (endometritis), should be conducted to discover the vast clinical application of secretome treatment <sup>[11]</sup>.

### 8.3 Regulatory Approval Status

Currently, no cell-based regenerative products derived from bovine secretome have received FDA or equivalent regulatory approval for veterinary use in the United States. International regulatory frameworks such as APQA, CVM and CVMP for stem cell products remain under development, with different jurisdictions maintaining varying requirements for demonstrating safety and efficacy <sup>[23]</sup>.

### 8.4 Batch-to-batch Consistency in Manufacturing

Robust standardisation of manufacturing protocols is an important requirement in regulatory approval, ensuring batch-to-batch consistency of secretome production. The variability introduced by different bovine tissue sources of cells, culture conditions and preparation methodologies must be minimised or control in order to maintain uniformity of the product <sup>[2]</sup>.

## 9. Future Directions and Research Priorities

### 9.1 Systematic Clinical Trials

For successful clinical translations, randomised controlled studies comparing the secretome efficacy to the current available treatment options. Such trials should assess primary outcome measures such as disease resolution, milk production recovery and reproductive efficiency while monitoring safety parameters <sup>[11]</sup>.

### 9.2 Clarification on the Mechanism and Interaction of the Components

Further research should elucidate the contribution of EV and soluble factors-mediated therapeutic effects and the interaction between various bioactive components in the secretome. The clarity in the mechanism of action of the secretome will be understood when particularly targeting the synergistic interactions between the bioactive components and interactions between the secretome and the organism <sup>[21, 22]</sup>.

### 9.3 Optimisation and Stimulation of Secretome Production

Investigating the pre-conditioning strategies of bovine MSCs, including hypoxia and inflammatory priming, which were observed to enhance the therapeutic properties of the derived secretome, could improve the efficacy for a particular application <sup>[4]</sup>. For regulatory approval and widespread clinical adoption, methodologies that are economically suitable for implementation and development of simplified and reproducible techniques are essential <sup>[2]</sup>.

### 9.4 Delivery System Development

The exploration of biomaterial carriers and targeted delivery mechanisms to improve the *in vivo* retention, cellular uptake and absorption of secretome components could enhance the therapeutic efficacy. This approach was studied by Wihadmadayatami *et al* <sup>[24]</sup> as the bovine umbilical mesenchymal stem cell-derived secretome, coupled with chitosan-hydrogel nanoparticles, was observed to be a promising approach to treat neurodegenerative conditions.

## 10. Conclusion

The secretome derived from the bovine mesenchymal stem cells offers a promising cell-free therapeutic strategy for addressing economically significant diseases in cattle. It contains a diverse array of bioactive compounds, including growth factors and immunomodulatory molecules, antimicrobial peptides and extracellular vesicles. These components collectively promote tissue regeneration, suppress inflammation, enhances angiogenesis and exert antimicrobial effects.

The therapeutic applications in mastitis, endometritis and wound healing have demonstrated clinical promise and the mechanism of action has been substantially elucidated through *in vitro* and limited *in vivo* studies. The cell-free nature of this approach provides distinct advantages in manufacturing, safety and logistics compared to direct MSC transplantation and enables large-scale production from accessible bovine tissue sources.

However, translation to routine clinical practice requires further development, including systematic clinical trials, standardisation manufacturing protocols, a regulatory approval framework and clarification of the mechanisms underlying EV-mediated versus soluble factor-mediated effects.

The substantial economic burden of mastitis, endometritis and other productivity-limiting conditions in cattle highlights the need to develop advanced bovine secretome therapeutics for future clinical implementation, with potential secondary benefits for human disease models through translational research.

### Conflict of interest

No conflict of interest

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